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🚀 RNN & LSTM Internal Hackathon | Smart Energy Consumption ForecastingCan we use the previous 24 hours of energy usage to predict what a household or building will consume in the next hour?That was the problem I explored during a Deep Learning Hackathon, where I built a time-series forecasting model using LSTM (Long Short-Term Memory). 📊 Dataset amp; Preprocessing 🔹 720,000 records across multiple entities 🔹 23 columns with 21 input features 🔹 Previous 24 hours used to predict the next hour 🔹 Cyclical encoding for hour-of-day 🔹 StandardScaler for feature scaling 🔹 Chronological 80/20 train-test split per entity to avoid data leakageI started with a SimpleRNN baseline and then moved to LSTM, which is better suited for capturing longer-term dependencies in sequential data. 🧠 Final LSTM ArchitectureLSTM (64) → Dropout (0.3) → Dense (32) → OutputThe model contained 24,129 trainable parameters and was trained for 30 epochs using Adam optimization with MSE loss and MAE as the evaluation metric.One of the most interesting parts was understanding the models learning behaviour. 📉 Training loss continued to decrease as the model learned the underlying patterns. 📈 However, validation loss began increasing after the early epochs, indicating overfitting.The actual-vs-predicted analysis showed that the model captured the overall energy consumption trend, although some higher-consumption values were more difficult to predict.A few test predictions: 🔹 Actual: 2.15 kWh → Predicted: 2.40 kWh 🔹 Actual: 2.76 kWh → Predicted: 2.47 kWh 🔹 Actual: 2.81 kWh → Predicted: 2.45 kWh 🔹 Actual: 5.57 kWh → Predicted: 5.77 kWhThe biggest takeaway was that time-series forecasting is not simply about reducing training loss. It requires careful attention to temporal ordering, sequence construction, data leakage, feature scaling, model selection, and generalization.Moving from SimpleRNN → LSTM + Dropout helped me understand how model architecture and regularization can improve the learning of temporal patterns.This hackathon gave me valuable hands-on experience applying Deep Learning to a real-world sustainability problem, with potential applications in smart buildings, energy management, demand forecasting, and efficient resource utilization.A special thanks to my mentor, Mayura Nagar, for the guidance, support, and valuable feedback throughout this project. Your mentorship played an important role in making this learning experience meaningful. 🙌

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